

Mini Project -3

Report: Human Activity Recognition using Neural Networks

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1-Abstract

Human Activity Recognition (HAR) aims to identify different activities based on sensor data. This project explores HAR using a dataset collected from a smartphone's accelerometer and gyroscope. The neural network approach is employed to classify activities like walking, sitting, standing, etc. The report presents findings from varying network configurations and the impact of dropout on model performance.

2- Introduction

Human Activity Recognition (HAR) involves analyzing sensor data to classify activities. This project utilizes data from a Samsung Galaxy S II smartphone's accelerometer and gyroscope, capturing 3-axial linear acceleration and angular velocity signals. The objective is to classify six human activities. Class 1 as downstairs motion, Class 2 as jogging motion, Class 3 as sitting motion, Class 4 as standing motion, Class 5 as upstairs motion, Class 6 as walking motion. The neural network architecture is implemented for this classification task.

3-Terminologies

3.1- Human Activity Recognition (HAR):

HAR refers to the process of automatically identifying and classifying human activities from sensor data. It involves recognizing activities like walking, running, sitting, or standing based on patterns detected in the sensor data.

3.2- Accelerometer:

An accelerometer is a sensor that measures proper acceleration, which is the rate of change of velocity with respect to time. It is used to sense the magnitude and direction of the acceleration vector.

3.3- Gyroscope:

Gyroscope is a sensor that measures the rate of rotation or angular velocity around a particular axis. It provides information about the orientation and rotational movement of an object.

3.4- Forward Pass:

The forward pass is the phase in a neural network where input data is fed into the network, and computations are performed layer by layer until the output is generated. It involves calculating the output of the neural network based on the given input data and the current set of weights.

3.5- Backward Pass (Backpropagation):

The backward pass, also known as backpropagation, is the phase in neural network training where the gradient of the loss function with respect to the weights and biases of the network is calculated. It's an essential step for adjusting the weights to minimize the loss during the training process.

3.6- Momentum:

Momentum is a technique used in optimization algorithms, particularly gradient descent, to speed up convergence and prevent oscillations during training.

3.7- Dropout:

Dropout is a regularization technique used to prevent overfitting in neural networks by randomly deactivating a fraction of neurons during training.

4- Activation Functions:

4.1- ReLU (Rectified Linear Unit): It is an activation function used in neural networks, efficiently addresses the vanishing gradient problem and accelerates convergence during training by introducing non-linearity to the network.

4.2- Softmax: It's an activation function used in multi-class classification problems in the output layer. It normalizes the output into a probability distribution over multiple class. Softmax ensures that the sum of output values across all classes equals 1, allowing the model to predict the class with the highest probability.

4.3- Categorical Cross-Entropy Loss: Categorical Cross-Entropy is a loss function used in multi-class classification problems to measure the dissimilarity between predicted and actual probability distributions.

5- Dataset:

The dataset comprises 7352 training and 2947 test patterns, capturing signals from different human activities. These time domain signals are obtained from the smartphone's sensors and processed to derive a 561-dimensional feature vector per pattern. These features include statistical measures like mean, standard deviation, variance, etc.

6- Methodology

6.1. Neural Network Architecture:

Layer Configuration: The neural network comprises an input layer, two hidden layers ($N_1 = 300$, $N_2 = 100, 200$), and an output layer.

Activation Functions:

ReLU for hidden layers: Introduces non-linearity efficiently by replacing negative values with zero.

Softmax for output layer: Computes class probabilities for multi-class classification tasks.

6.2. Training Setup:

Parameter Initialization: Weights and biases initialized randomly using a uniform distribution in the range $[-0.1, 0.1]$.

Backpropagation Algorithm: Utilization of backpropagation to update parameters, employing gradient descent with momentum for optimization.

Hyperparameters:

Learning rate (0.001, 0.01): Controls the step size in the parameter update process.

Momentum (0, 0.9): Governs the effect of the previous update in the parameter update process.

Batch size (0, 50): Defines the number of samples processed before updating the model.

Validation Set and Early Stopping: Creation of a validation set (10% of training data) for monitoring model performance and early stopping based on validation accuracy to prevent overfitting.

6.3 Dropout Regularization:

Application of Dropout: Employing a dropout rate of 0.5 specifically to the second hidden layer to prevent overfitting by randomly deactivating neurons during training.

Integration in Training: Incorporation of dropout in the training process, considering both online learning and mini-batch processing scenarios.

6.4. Training Loop and Reporting:

Epoch Iteration: Iterating through epochs and monitoring key metrics like validation accuracy and training error.

Misclassification Analysis: Identifying and analyzing misclassified patterns in both training and test sets for result analysis and model enhancement.

7- Results:

7.1- Configuration Evaluation:

Learning Rate: Tested with 0.001 and 0.01.

Hidden Layer Sizes: N1=300, N2=100,200.

Momentum: Examined at 0 and 0.9.

Batch Size: Explored with 0 and 50.

7.2- Performance

The table displays the test accuracy for different configurations of hyperparameters in a neural network model.

Analyzed implementation of early stopping in training the neural network.

Learning Rate	N1	N2	Momentum	Batch Size	Accuracy
0.01	300	100	0	0	0.718
				50	0.777
			0.9	0	0.1598
				50	0.167
		200	0	0	0.6952
				50	0.783
			0.9	0	0.1598
				50	0.167
0.001	300	100	0	0	0.7475
				50	0.7051
			0.9	0	0.1598
				50	0.167
		200	0	0	0.743
				50	0.661
			0.9	0	0.1598
				50	0.167

7.1- Result Analysis:

Different configurations of hyperparameters (Learning Rate, N1, N2, Momentum, Batch Size), and the accuracy values reflect the performance of the model under each configuration

Insights:

Batch Size Impact: Generally, Batch Size 50 seems more beneficial in achieving higher accuracies compared to Batch Size 0 across different hyperparameter configurations.

Learning Rate Influence: Learning Rate 0.01 tends to produce better accuracies compared to 0.001 in various settings.

Layer Sizes (N1, N2): The configuration with N1=300 and N2=100 shows better performance compared to N1=200, especially when paired with a Batch Size of 50.

9- Conclusion

The experimentation with varied neural network configurations for HAR showed distinct effects on model performance. Hyperparameters, emphasizing the importance of Batch Size, Learning Rate, and Layer Sizes in determining the success of the neural network model for Human Activity Recognition. Integrating early stopping and dropout techniques further boosted the model's performance and generalization capabilities.